# ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S [ANSWER KEY]

## Unit 8.01: Power Series Solutions & Method of Frobenius

**Q1.** For x^{2} - xy''+y'-y=0 , the point x=0 is:

[ ] A: ordinary point
[ ] B: a point where no solution exists
**[x] C: singular point**
[ ] D: a point where every solution vanishe

Solution: Dividing by x^{2} - x gives y''+Pxy'+Qxy=0 . The leading coefficient vanishes at x=0 , so P,Q are not analytic there: it is a singular point.

**Q2.** Near x_0=0 , the power series solutions of x + 4y''+xy'+y=0 are guaranteed to converge at least for |x-x_0|<R with R=

[ ] A: 5
**[x] B: 4**
[ ] C: infty
[ ] D: 1

Solution: The singular points are the zeros of x + 4 : -4 complex ones count. The nearest is at distance 4 from 0 .

**Q3.** For x^{2} - 9y''+y'-y=0 , the point x=-1 is:

[ ] A: singular point
**[x] B: ordinary point**
[ ] C: a point where no solution exists
[ ] D: a point where every solution vanishe

Solution: Dividing by x^{2} - 9 gives y''+Pxy'+Qxy=0 . P and Q are analytic at -1 iff the leading coefficient is nonzero there; x^{2} - 9=-8 , so it is an ordinary point.

**Q4.** Near x_0=0 , the power series solutions of x^{2} - 4 x + 3y''+xy'+y=0 are guaranteed to converge at least for |x-x_0|<R with R=

[ ] A: 3
[ ] B: infty
**[x] C: 1**
[ ] D: 2

Solution: The singular points are the zeros of x^{2} - 4 x + 3 : 1, 3 complex ones count. The nearest is at distance 1 from 0 .

**Q5.** For x^{2} + xy''+y'-y=0 , the point x=-1 is:

**[x] A: singular point**
[ ] B: ordinary point
[ ] C: a point where no solution exists
[ ] D: a point where every solution vanishe

Solution: Dividing by x^{2} + x gives y''+Pxy'+Qxy=0 . The leading coefficient vanishes at x=-1 , so P,Q are not analytic there: it is a singular point.

**Q6.** Near x_0=0 , the power series solutions of x^{2} + 4y''+xy'+y=0 are guaranteed to converge at least for |x-x_0|<R with R=

**[x] A: 2**
[ ] B: infty
[ ] C: 1
[ ] D: 4

Solution: The singular points are the zeros of x^{2} + 4 : - 2 i, 2 i complex ones count. The nearest is at distance 2 from 0 .

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**Q7.** For x^{2} - 4 xy''+y'-y=0 , the point x=0 is:

[ ] A: ordinary point

[ ] B: a point where no solution exists

[ ] C: a point where every solution vanishe

**[x] D: singular point**

Solution: Dividing by x^{2} - 4 x gives y''+Pxy'+Qxy=0 . The leading coefficient vanishes at x=0 , so P,Q are not analytic there: it is a singular point.

**Q8.** Near x_0=0 , the power series solutions of x^{2} + x - 6y''+xy'+y=0 are guaranteed to converge at least for |x-x_0|<R with R=

**[x] A: 2**

[ ] B: 3

[ ] C: infty

[ ] D: 1

Solution: The singular points are the zeros of x^{2} + x - 6 : -3, 2 complex ones count. The nearest is at distance 2 from 0 .